

SUPPLEMENTAL MATERIAL B: KDUQ, CHUQ, AND TEST CORPORA

In this section, we list the experimental data used to train and test the UQ OMPs described in the main text. To enable direct comparison with the original KD and CH89 treatments, we re-assembled (as faithfully as possible) the same corpora of experimental training data listed in the original publications, which we refer to the “KD corpus” and “CH89 corpus”, respectively. To distinguish these corpora as listed in the original publications from our present reconstruction of these corpora, we use the names “KDUQ corpus” and “CHUQ corpus” to refer to the corpora codified in this section. Also tabulated are the data comprising the “Test corpus”, all of which post-date the publication of the original CH89 [19] and KD [20] treatments (see main text for bibliography).

Before detailing the assembly of these corpora, a few methodological comments are worth making. Wherever possible, for the data sets listed in the KD and CH89 corpora, we drew on experimental data as tabulated in the EXFOR experimental reaction database [31]. In the instances where we could not locate the data in EXFOR, we turned to the original publication (as listed in the canonical KD or CH89 papers) and retrieved the data from that source. In the few remaining cases where the data either could not be found in the original literature or were missing critical features (such as experimental errors), we omitted the data set from our reconstructed corpora and documented the discrepancy. For consistency, we prioritized the labels and metadata from the EXFOR entries, as they existed at the time of our accession of those data. If a data set was unavailable or incomplete in EXFOR, we then used the labels and metadata as we interpreted from the original literature source. We justify this as an attempt to make our analysis as reproducible as possible: because EXFOR is regularly updated as errors are identified and corrected, our hope is that future investigators wishing to use the same data corpus need only refer to the EXFOR accession numbers we provide here and not start “from scratch”. A consequence of this approach is that our reconstructed corpora span slightly fewer data compared to the the original corpora because we were unable to locate, either in EXFOR or in the original literature, some data sets that were nominally included in the original corpus. These few instances we attribute either to typographic errors in the references as listed in the KD and CH89 publications or our own error in searching the available literature. For construction of the Test corpus, all data were drawn from EXFOR.

Once the data sets were collected and metadata assigned, we applied a series of munging steps to prepare the corpus for OMP analysis. First, units were homogenized: scattering energies, scattering angles, integral cross sections, and differential cross sections were transformed to MeV (lab frame), degrees (center-of-momentum frame), mb, and mb/sr, respectively. All pro-

ton differential elastic cross section data sets that were provided relative to Rutherford scattering were rescaled into absolute terms (assuming relativistic kinematics). Any datum that was missing a “necessary feature” was flagged and removed from the KDUQ/CHUQ corpora. For example, for differential elastic cross sections, these necessary features were: scattering energy, scattering angle, cross section, and cross section error. Many EXFOR-based data list more than one type of error, for instance, digitization error, statistical error, etc. For a given datum, we assigned the overall error according to the following list of EXFOR error labels, in order of preference: ERR-T (total error), (+DATA-ERR + -DATA-ERR)/2 (average of positive and negative error), ERR-S (statistical error), ERR-DIG (digitization error), ERR-SYS (systematic error). Absent a covariance matrix for each data set (none were available for the data sets in the KDUQ/CHUQ corpus), our assignment of overall error is not unique, but is at least reproducible. Additionally, in the KD corpus, many of the neutron total cross section data sets had an unnecessarily large number of energy bins for the requirements of optical potential optimization. For instance, the data set from 250 keV to 20 MeV for ^{27}Al (EXFOR Acc. No. 22331004) includes nearly 50,000 unique energy bins. Calculating cross sections for each of these energies in our analysis would be computationally expensive for negligible benefit, given the slowly-varying nature of the total cross section in this energy regime. For these neutron total cross section data sets, rather than include the full complement of neutron total cross section data, we downsampled each data set to have one datum per MeV — an energy step still sufficiently fine under the smoothness assumptions inherent to a global optical potential.

Due to rounding and discrepancies in tabulation, the scattering energies listed here do not always exactly match those reported from the KD corpus or the references therein. The scattering energies listed here are as provided in EXFOR, or, if not available there, transcribed directly from the literature reference. To facilitate others’ use of the KDUQ corpus for future optical model work, we list the EXFOR accession number the data sets that we drew from EXFOR. For the few data sets that were not available in EXFOR, this field is filled with “-” in the following tables, and the original reference listed in the Comments following each table. The Comments also document special cases and notable differences between the original corpora and our reconstructions. In the tables, all listed energies are in units of MeV.

Assembling the KDUQ corpus

Tables 1, 2, and 7 of [20] list five of the six sectors of experimental data used in the original Koning-Delaroche analysis: neutron differential elastic scattering cross sections and analyzing powers, neutron total cross sections,

and proton differential elastic scattering cross sections and analyzing powers. The sixth sector, proton reaction cross sections, was drawn from [71] and [72], following the original analysis.

During data assembly we noted two general types of discrepancy between the labeling and citations given in the original Koning-Delaroche analysis, the data currently available in EXFOR, and the labeling as reported in the original experimental literature. The first type of discrepancy was a difference in target nucleus listed for a given data set. In some cases, this involved a data set being labeled as being for an elemental (natural abundance) target in one source but for an isotopic target in another (e.g., ^{63}Cu in EXFOR and the original reference, but $^{\text{nat}}\text{Cu}$ in the KD corpus). In one case, the KD corpus lists a neutron differential elastic scattering data set on $^{\text{nat}}\text{La}$ at 7.8 MeV, but the cited reference lists the scattering target as $^{\text{nat}}\text{Al}$, likely a typographic error. Because both the target A and asymmetry $(N - Z)/A$ enter the potential definition, errors in these metadata could bias the potential parameters, especially those with exponential dependence on A . The second type of discrepancy concerned several data sets listed in the original KD corpus as having scattering energies we were unable to locate either in EXFOR or in the original literature. In a few of these cases, the issue was inaccessibility of the original literature source (e.g., unpublished work, conference proceedings, or older PhD theses). In other cases, we located experimental data in the original source, but could not find data points or sets at the specified scattering energies. These data sets were omitted from the KDUQ corpus. While the number of data sets affected by this issue was small, because many KD potential depth parameters depend on the scattering energy to the second and third power, any incorrect scattering energy labels could have significant effect unless an outlier procedure, such as that detailed in the main text, is applied.

Assembling the CHUQ corpus

Table 5 of [19] lists the four sectors of experimental data used in the original CH89 analysis: differential elastic scattering cross sections and analyzing powers for both neutrons and protons. Proton data were collected at four facilities: the Triangle Universities Nuclear Laboratory (TUNL), the Eindhoven University of Technology, Oak Ridge National Laboratory (ORNL), and the Research Center for Nuclear Physics (RCNP) in Osaka. Neutron data were collected at two facilities: TUNL and the Ohio University Accelerator Laboratory (OUAL). As the authors of CH89 point out, the fact that only a few laboratories were used reduces the likelihood of systematic discrepancies between data sets, a potential advantage for their OMP analysis. During assembly, the main discrepancy between the metadata as listed the original analysis and the data in EXFOR were a handful of miss-

ing (p,p) data sets on ^{58}Ni , ^{142}Nd , ^{148}Sm , and ^{150}Sm from Eindhoven. Neither EXFOR nor the primary literature cited for these data sets refer to these targets, or if the target is included, these sources give a different scattering energy for the target in question. We struck these data sets from the CHUQ corpus. Given that these data sets comprise a small fraction of the overall corpus, it is unlikely their omission impacted our analysis.

A more difficult obstacle was faithful assignment of experimental uncertainties. For thirty-six of the proton data sets and twelve of the neutron data sets used for CH89, the corresponding EXFOR entry included multiple columns of partial experimental uncertainty. For many proton data sets, one or more error columns consisted of almost all “null” values and no additional information was available on whether the multiple columns of experimental uncertainties were correlated or independent. For consistency across all data sets, we dropped entirely any mostly-“null” columns and accepted the uncertainties from the other column as the overall uncertainty. Consequently, for a few of the proton scattering data in the CHUQ corpus, the associated experimental uncertainties may differ than those used in the original CH89 corpus. In the 12 neutron data sets with multiple experimental uncertainty columns, neither column had “null” values. We chose to combine the columns in quadrature into a new “overall uncertainty” column. Such cases are documented in the notes following the CHUQ corpus tables below. Regardless of these potential differences, because we employed both outlier rejection and a residual unaccounted-for uncertainty term in our OMP characterizations, we expect that these differences had negligible impact on the present analysis.

Lastly, rather than fit the analyzing power data directly (as in Koning-Delaroche), the authors of CH89 chose to fit proton analyzing power data as “analyzing cross sections” $A_y \times \sigma_{el}$ (analyzing powers times elastic cross sections). They explained that “this observable is preferable to A_y , since theoretically A_y is computed from it by dividing by σ .” They estimated the uncertainty of $A_y \times \sigma_{el}$ by combining the individual uncertainties of A_y and σ_{el} according to uncorrelated propagation of uncertainties, which they noted was justified by a separate analysis. The practical effect was to give “polarization observables in the $A_y \times \sigma_{el}$ form... a more equitable weighting in the fitting process.” In our analysis, we chose to fit analyzing powers directly to avoid treating neutron and proton analyzing data differently. At low angles, the low-angle proton differential elastic cross sections are dominated by Coulomb and are often orders of magnitude larger than those cross sections at higher angles. Thus, low-angle proton cross sections typically have far smaller reported statistical uncertainties than neutron cross sections at the same angles. As such, it is possible that using “analyzing cross sections” could result in bias to asymmetry-dependent terms, so we elected to fit analyzing powers directly. However, due to

our employment of outlier-rejection and unaccounted-for uncertainties, our judgment was that treating the analyzing powers directly versus rescaling them into “analyzing cross sections” would not appreciably impact our results.

Assembling the Test corpus

Our general criteria for populating the Test corpus were as follows:

- Data should be directly measured, not a derived or calculated quantity
- Data collection should post-date the publication of the original KD and CH89 treatments
- Data should be of the same types (e.g., absolute neutron total cross sections, differential elastic cross sections) used in the training corpora
- Data should be plausibly describable by a low-to-intermediate-energy optical potential treatment (excluding, for example, scattering measurements at 1 GeV, or proton elastic scattering measurements probing resonances below the Coulomb barrier)
- Data should have the minimum essential features and meta-data enabling consistent treatment beside the CH89 and KD corpora, including experimental uncertainties, and be free of plain typographic or methodological issues

We canvassed the EXFOR database for data sets subject to the above constraints and recovered roughly 250 data sets. Among data fulfilling these criteria, several special cases were encountered; our treatment of these data is detailed in the Comments following each data table. Overall, we note that compared to the four decades prior to the publication of the KD global potential (1960-2003), the rate of scattering data measurements has declined in the last two decades, at least as indicated by the number of new EXFOR entries. As the reach and number of radioactive beam facilities expand, smaller cyclotron facilities are closed, digital data acquisition becomes the norm, and funding preferences change, the selection of scattering targets and energies has also shifted. Due to these factors it seems likely that the data comprising the Test Corpus represent a somewhat different underlying distribution than those comprising the KD and CH89 corpora; analysts wishing to compare the performance of their OMPs against these corpora should keep this in mind.

KDUQ CORPUS

Neutron differential elastic cross sections

Isotope	Energy	EXFOR	Acc.
^{24}Mg	3.4	30463008	
	6.1	21490002	
	9.76	21773002	
	14.83	21773010	
$^{\text{nat}}\text{Mg}$	1.969	11493004	
	2.958	11493004	
	3.948	11493004	
	4.958	11493004	
	5.44	10105006	
	6.44	10105006	
	7.55	10105006	
	7.97	11455002	
	8.56	10105006	
	11.01	10633002	
	14.1	11286009	
	21.6	22048002	
^{27}Al	3.2	11511002	
	5.44	10106025	
	6.44	10106025	
	7.54	10106025	
	7.62	22532002	
	8.56	10106025	
	10.873	12875002	
	13.883	12875002	
	16.883	12875002	
	18	12973005	
	20	12973005	
	22	12973005	
	25	12973005	
	26	12973005	
	84	11472002	
	96	21123003	
	136	21374007	
^{28}Si	21.7	13125002	
$^{\text{nat}}\text{Si}$	5.44	10107006	
	6.37	10107006	
	6.44	10107006	
	7.55	10107006	
	7.958	12927006	
	8.56	10107006	
	9.948	12927006	
	11	10697002	
	11.936	12927006	
	13.965	12927006	
	16.917	12927006	
	20	10697003	
	26	10697003	
	30.3	13902002	
	40	13902002	
	55	22480003	
	65	22480003	
	75	22480003	
^{31}P	3.5	20341010	
	3.85	20341010	
	4.2	20341010	
	4.5	20341010	
	4.8	20341010	
	5.95	21046008	
	7.79	10263006	
	9.05	10263006	
^{32}S	21.7	13125004	
$^{\text{nat}}\text{S}$	3	20019020	
	4	20019020	
	5.5	10108008	
	6.37	10108008	
	7.05	20019020	
	7.6	10108008	
	7.956	12927008	
	8.52	10108008	
	9.947	12927008	
	11.01	10633004	
	11.933	12927008	
	13.922	12927008	
	16.917	12927008	
	20	10697004	
	26	10697004	
	30.3	12871002	
	40.3	12871002	
$^{\text{nat}}\text{Cl}$	14.1	41003002	
$^{\text{nat}}\text{K}$	3	21098002	
	3.74	11618004	
	4.33	11618004	
	6.52	11618004	
	7.91	11618004	
$^{\text{nat}}\text{Ar}$	7.75	22121004	
	14	11418005	
^{40}Ca	2.06	11618009	
	3.29	11618009	
	5.3	11618009	
	5.88	11618009	
	6.52	11618009	
	7.91	11618009	
	9.91	12785002	
	11.01	10633005	
	11.91	12785002	
	13.9	12785002	
	16.916	12996002	
	19.0	13127002	
	20.0	10697005	
	21.6	22048010	
	26.0	10697005	
	30.3	12724002	
	40.0	12724002	

^{nat} Ca	65.0	13510004		11.93	10958002
⁴⁵ Sc	2.62	13547002		13.92	10958002
	2.86	13547002		16.929	12997002
	3.83	13547002		20	12862002
	5.0	13547003		24	12862002
	5.9	13547003		26	12862002
	6.5	13547003	⁵⁵ Mn	2.47	20019082
	7.14	13547003		3.0	20019082
	8.03	13547003		3.49	20019082
	9.06	13547003		4.0	20019082
	10.0	13547003		4.56	20019082
^{nat} Ti	4.5	13689002		6.09	20019082
	5.5	13689002		7.05	20019082
	6.5	13689002		8.05	20019082
	7.55	13689002		11.01	10633007
	8.08	13689002	^{nat} Fe	55.0	22480004
	8.41	13689002		65.0	22480004
	9.06	13689002		75.0	22480004
	9.5	13689002	⁵⁶ Fe	4.6	11708006
	9.99	13689002		5.0	11708006
	13.95	11486011		5.05	10037024
⁵¹ V	5.44	10110002		5.56	11708006
	6.44	10110002		6.53	11708006
	7.99	22409002		7.55	11708006
	8.56	10110002		7.96	10958004
	9.01	22409002		9.94	10958004
	10.07	22409002		11	12862003
	10.91	22409002		11.93	10958004
	11.9	22409002		13.92	10958004
	13.03	22409002		20	12862003
	14.37	22409002		21.6	22048014
⁵² Cr	3.0	20019063		24.8	11490003
	4.0	20019063		26	12862003
	4.34	10413002	⁵⁸ Ni	7.904	12930002
	4.92	10413002		9.958	12930002
	6.44	10413002		11.952	12930002
	7.54	10413002		13.941	12930002
	7.95	22408002		16.934	12997006
	8.56	10413002		24.0	10953002
	9.0	22408002	^{nat} Ni	3.0	20019132
	9.8	22408002		3.49	20019132
	10.79	22408002		4.0	20019132
	11.44	22408002		4.56	20019132
	12.01	22408002		4.92	10113002
	12.7	22408002		6.09	20019132
	13.65	22408002		6.44	10113002
	14.1	22408002		7.05	20019132
	14.76	22408002		7.54	10113002
	21.6	22048012		11.01	10633010
⁵⁴ Fe	7.0	10469002		21.6	22048018
	8.5	10469002	⁵⁹ Co	2.0	20019112
	9.94	10958002		2.47	20019112
	11	12862002		3.0	20019112
				3.49	20019112

	4.0	20019112		⁸⁸ Sr	11.0	10729004
	4.56	20019112		⁸⁹ Y	3.83	12835003
	5.0	10112012			4.5	12979002
	5.44	10112012			5.0	12979002
	6.09	20019112			5.5	10130002
	6.44	10112012			5.9	12979002
	7.05	20019112			6.5	12979002
	7.55	10112012			7.14	12979002
	8.05	20019112			7.5	12979002
	8.56	10112012			7.962	12994002
	11.01	10633009			8.4	12979002
	13.95	11486015			9.06	12979002
	21.6	22048016			9.5	12979002
	23.0	12962004			9.954	12994002
⁶⁰ Ni	4.34	10113007			11.0	12774002
	4.92	10113007			11.944	12994002
	6.44	10487002			13.934	12994002
	7.54	10487002			16.928	12994002
	7.904	12930004			21.6	22048020
	8.56	10487002		⁹⁰ Zr	2.0	10468003
	9.958	12930004			2.11	10621002
	11.952	12930004			2.6	10468003
	13.941	12930004			3.0	10468003
	24.0	10953004			3.5	10468003
⁶³ Cu	5.5	10414002			4.0	10468003
	7.0	10414002			5.17	10621002
	7.96	10958006			5.9	21638002
	8.5	10414002			6.95	21638002
	9.94	10958006			7.75	21638002
	11.93	10958006			8.0	13160002
	13.92	10958006			10.0	13160003
^{nat} Cu	1.6	12869003			11.0	10729002
	2.0	12869003			24.0	13160004
	2.2	12869003		⁹¹ Zr	8.0	13160007
	2.6	12869003			10.0	13160008
	3.0	12869003			24.0	13160009
	3.4	12869003		^{nat} Zr	55.0	22480005
	3.8	12869003			65.0	22480005
	84.0	11472003			75.0	22480005
	96.0	21123006		⁹² Mo	1.5	10391018
	155	-			1.8	10524002
^{nat} Ge	7.55	11202016			2.0	10524002
^{nat} As	8.05	20162006			2.2	10524002
^{nat} Se	1.0	11637022			2.4	10524002
	3.2	11511017			2.6	10524002
	3.66	11617019			2.8	10524002
	14.1	20199003			3.0	10524002
⁸⁰ Se	4.0	40221014			3.2	10524002
	8.0	12887003			3.4	10524002
	10.0	12887003			3.6	10524002
^{nat} Sr	3.2	11511018			3.8	10524002
	3.66	11617024			4.0	10524002
	4.37	40101008			6.0	12782002
	14.76	20223006				

	9.0	10867002		2.0	10524004
	11.0	10867002		2.2	10524004
	20.0	10867002		2.4	10524004
	26.0	10867002		2.6	10524004
⁹² Zr	2.0	10468006		2.8	10524004
	3.0	10468006		3.0	10524004
	4.0	10468006		3.2	10524004
	5.9	21638004		3.4	10524004
	6.95	21638004		3.6	10524004
	7.75	21638004		3.8	10524004
	8.0	13160012		4.0	10524004
	10.0	13160013		9.0	10867004
	24.0	13160014		11.0	10867004
⁹³ Nb	2.55	12797003		20.0	10867004
	2.85	12797003		26.0	10867004
	4.5	12937002	¹⁰⁰ Mo	1.5	10391028
	5.0	12937002		1.8	10524005
	5.5	12937002		2.0	10524005
	5.9	12937002		2.2	10524005
	6.5	12937002		2.4	10524005
	7.14	12937002		2.6	10524005
	7.5	12937002		2.8	10524005
	8.03	12937002		3.0	10524005
	8.4	12937002		3.2	10524005
	9.06	12937002		3.4	10524005
	9.941	12995002		3.6	10524005
	11.01	10633011		3.8	10524005
	11.928	12995002		4.0	10524005
	13.915	12995002		6.0	12782014
	16.91	12995002		9.0	10867005
	20.0	13532002		11.0	10867005
⁹⁴ Zr	1.5	10391010		20.0	10867005
	8.0	13160017		26.0	10867005
	10.0	13160018	¹⁰³ Rh	1.5	12796003
	24.0	13160019		1.8	12796003
⁹⁶ Mo	1.5	10391023		2.3	12796003
	1.8	10524003		2.8	12796003
	2.0	10524003		3.25	12796004
	2.2	10524003		3.75	12796004
	2.4	10524003		4.505	13589002
	2.6	10524003		5.005	13589002
	2.8	10524003		5.505	13589002
	3.0	10524003		5.905	13589002
	3.2	10524003		6.505	13589002
	3.4	10524003		7.14	13589002
	3.6	10524003		7.505	13589002
	3.8	10524003		8.03	13589002
	4.0	10524003		8.405	13589002
	6.0	12782010		9.06	13589002
	9.0	10867003		9.505	13589002
	11.0	10867003		9.995	13589002
	20.0	10867003	^{nat} Pd	1.5	12757003
	26.0	10867003		1.8	12757003
⁹⁸ Mo	1.8	10524004		2.3	12757003

	2.8	12757003		24.0	10817006
	3.25	12757003	¹¹⁸ Sn	11.0	10817007
	3.75	12757003		24.0	10817007
	5.9	13182002	^{nat} Sn	6.04	11287037
	7.14	13182002		24.0	11490004
	8.03	13182002		65.0	13510006
¹⁰⁷ Ag	1.6	10876003	¹²⁰ Sn	1.55	12795002
	2.2	10876003		2.03	12795002
	2.8	10876003		2.15	12795002
	3.4	10876003		2.25	12795002
	4.0	10876003		2.45	12795002
^{nat} Ag	4.51	13638002		2.65	12795002
	5.01	13638002		2.75	12795002
	5.51	13638002		3.0	12795003
	5.91	13638002		3.4	12795003
	6.51	13638002		3.8	12795003
	7.14	13638002		4.0	12795003
	7.51	13638002		9.943	13158007
	8.03	13638002		13.923	13158007
	8.41	13638002		16.905	13158007
	9.06	13638002	¹²³ Sb	1.55	12805003
	9.51	13638002		2.05	12805003
	9.99	13638002		2.55	12805003
^{nat} Cd	2.25	12804002		3.0	12805004
	2.85	12804002		3.4	12805004
	3.4	12804003		4.0	12805004
	4.0	12804003		4.5	13802002
	4.5	13585002		5.5	13802002
	5.0	13585002		5.9	13802002
	5.5	13585002		6.5	13802002
	5.9	13585002		7.14	13802002
	6.5	13585002		7.55	13802002
	7.14	13585002		8.08	13802002
	7.5	13585003		8.41	13802002
	8.08	13585002		9.06	13802002
	8.41	13585002		9.5	13802002
	9.06	13585002		9.99	13802002
	9.5	13585002		14.0	11806004
	9.99	13585002	¹²⁴ Sn	11.0	10817010
	14.6	11465007		24.0	10817010
	96.0	21123007	¹²⁷ I	4.0	40221029
^{nat} In	4.5	13193002		16.1	21664012
	5.0	13193002	^{nat} Te	3.2	11511027
	5.9	13193002		14.0	40793008
	7.14	13193002	^{nat} Ba	1.0	11637055
	8.03	13193002		4.1	11621023
	9.06	13193002		5.0	11321009
	9.5	13193002	^{nat} La	2.545	20337002
	9.99	13193002		3.07	20337002
	11.01	10633016		3.578	20337002
	14.0	11568004	^{nat} Ce	0.98	21110007
¹¹⁶ Sn	9.945	13158005		3.2	11511030
	11.0	10817006		5.0	11877006
	13.925	13158005		21.6	22048022

¹⁴¹ Pr	1.2	21292003
	1.5	20337008
	2.03	20337008
	2.545	20337008
	3.07	20337008
	3.2	11511031
	3.578	20337008
	5.0	11877007
	8.0	12054005
¹⁴² Nd	7.0	21338002
¹⁴⁴ Nd	7.0	21338003
^{nat} Nd	0.98	21110009
¹⁴⁸ Sm	2.47	10718002
	6.25	10528002
	7.0	21337002
¹⁹⁷ Au	4.1	11215049
	5.0	11877009
	7.0	20346018
	8.05	20162024
^{nat} Hg	3.0	22029003
	3.2	11511034
	14.1	20199006
	16.1	21664014
²⁰⁶ Pb	7.0	20346025
	7.71	22121012
	8.05	20162030
	11.01	10633020
	13.7	40288002
	21.6	22048024
^{nat} Pb	155	-
²⁰⁸ Pb	1.8	400750041
	4.0	12903002
	4.5	12903002
	5.0	12903002
	5.5	12903002
	7.97	13531002
	9.0	10871002
	11.0	10871002
	13.9	13685002
	16.9	13685002
	20.0	10871002
	22.0	12865002
	23.0	12962006
	24.0	12865002
	26.0	10871002
	30.3	12701005
	40	12701005
	55.0	22480006
	65.0	22480006
	75.0	22480006
	84	11472004
	96.0	21123008
	136.0	21374010

²⁰⁹ Bi	2	10846003
	2.5	10846003
	3.0	22029006
	3	10846003
	3.53	10846003
	4	12903003
	4.5	12903003
	5	12903003
	5.5	12903003
	6	12903003
	6.5	12903003
	7	12903003
	7.5	13199003
	7.82	22121014
	8	13199003
	9	13199003
	10	13199003
	11	13199003
	12	13199003
	20	13199003
	21.6	22048026
	24	13199003

Comments

²⁸Si: The data sets at 30.3 and 40 MeV from (Phys. Rev. C 28 p.2530 (1983)) listed as being for ²⁸Si are actually from ^{nat}Si; we assigned them to ^{nat}Si for the KDUQ corpus.

^{nat}Si: There are 2 unique datasets at 6.4 MeV from ORNL Report 4517 (1970); which was used in the KD corpus is not specified. We kept both in the KDUQ corpus.

⁴⁰Ca: The data set at 11.0 MeV from NP/A 286 (1977) 232 was not available in EXFOR. In original literature, it was presented in a figure that was difficult to digitize; we omitted this dataset from the KDUQ corpus.

^{nat}Cu: We were unable to locate the data set at 155 MeV in EXFOR. For these data we used the same reference as listed in the KD corpus: R. S. Harding (Physical Review, Vol.111, p.1164 (1958)).

^{nat}Sr: The data set at 0.9 MeV from Cox and Dowing (ANL Report No. ANL-7935 (1972)) includes no errors; we omitted this dataset from the KDUQ corpus.

⁸⁹Y: The data set at 3.8 MeV is apparently from Budtz-Jorgensen et al. (Zeitschrift für Physik A, Hadrons and Nuclei, Vol.319, p.47 (1984)), not PRC 34 (1986) 1599 as listed in the KD corpus.

^{93}Nb : The data sets 2.6 and 2.9 MeV ascribed to Smith, Guenther, and Whalen (Nucl. Phys. A vol 415 issue 1, page 1 (1984) are apparently available only in an earlier, separate report by the same authors (ANL Report No. 70 (1982)). For the KDUQ corpus, we used the data as listed in the earlier report. Additionally, the data set at 14.6 MeV does not appear to exist in the reference cited by the KD corpus (Pedroni et al. (PRC 43 2336 (1991))); we omitted this dataset from the KDUQ corpus.

$^{\text{nat}}\text{Cd}$: The data set at 7.5 MeV from NP/A 568 221 (1994) is actually the sum of elastic scattering and inelastic scattering to the first excited state (this state is low-lying and could not be resolved from the ground state in the scattering experiment). This quantity was described as “pseudo-elastic” and apparently analyzed as elastic in the original KD corpus. For consistency, we included it in the KDUQ corpus.

^{120}Sn : The data set at 6.04 MeV from Wilenzick et al., Nucl. Phys. vol. 62, p. 511 (1965) listed as being for ^{120}Sn is actually from $^{\text{nat}}\text{Sn}$; we assigned them to $^{\text{nat}}\text{Sn}$ for the KDUQ corpus. The data set at 11 MeV listed in the KD Corpus as being for ^{120}Sn apparently does not exist in Guss et al. (PRC 39 (1989) 405); we omitted this dataset from the KDUQ corpus.

$^{\text{nat}}\text{La}$: The data set at 7.8 MeV is listed in the KD corpus as being from Dagge et al. (PRC 39 (1989) 1768), but this reference reports scattering measurements on ^{27}Al , not $^{\text{nat}}\text{La}$. As we were unable to locate any data set in EXFOR at this scattering energy for $^{\text{nat}}\text{La}$ that might correspond to the experimental data shown in Fig. 28 of the canonical KD treatment, we omitted this dataset from the KDUQ corpus.

^{142}Nd : The data set at 2.5 MeV from Bull. Am. Phys. Soc. 24 (1979) 854 includes no errors; we omitted this dataset from the KDUQ corpus.

^{144}Nd : The data set at 2.5 MeV from Bull. Am. Phys. Soc. 24 (1979) 854 includes no errors; we omitted this dataset from the KDUQ corpus.

^{197}Au : The data set at 2.5 MeV from Day (Priv. comm. (1965)) includes no errors; we omitted this dataset from the KDUQ corpus.

$^{\text{nat}}\text{Hg}$: The data set listed in the KD corpus as having a scattering energy of 14.8 MeV from Nauta et al. (Nucl. Phys. 2 (1956) 124) is actually at 14.1 MeV, as reported in the original publication. We assigned a scattering energy of 14.1 MeV for the KDUQ corpus.

$^{\text{nat}}\text{Pb}$: We were unable to locate the data set at 155 MeV in EXFOR. For these data we used the same reference as listed in the KD corpus: R. S. Harding (Physical Review, Vol.111, p.1164 (1958)).

^{208}Pb : The data set at 21.6 MeV from Olsson et al. (NP/A 472 (1987) 237) was apparently from a target 88.2% enriched in ^{206}Pb , and already appears as a dataset in the KD corpus under ^{206}Pb . As such, we omitted it from ^{208}Pb in the KDUQ corpus. The data set at 155 MeV from Phys. Rev. 111 (1958) 1164 listed in the KD corpus as being for ^{208}Pb is actually from $^{\text{nat}}\text{Pb}$; we assigned it to $^{\text{nat}}\text{Pb}$ for the KDUQ corpus.

^{209}Bi : For the data sets from 2-3.53 MeV from Das and Finlay (PRC 42 (1990) 1013), we used data from Nucl. Sci. Engrg. 75 69 (1980) (reference 220 in the KD Corpus) for the KDUQ corpus. For the data sets from 4.0-7.0 MeV from Das and Finlay (PRC 42 (1990) 1013), we used data from NP/A 443 249 (1985) (reference 23 in the KD Corpus) for the KDUQ corpus.

Neutron analyzing powers

Isotope	Energy	EXFOR Acc.
²⁷ Al	7.62	22532004
	14	13684002
	17	13684002
⁴⁰ Ca	9.91	12785003
	11.91	12785003
	13.9	12785003
	16.923	12996004
⁵⁴ Fe	9.941	12997004
	13.937	12997004
	16.93	12997004
⁵⁸ Ni	9.906	12997008
	13.94	12997008
	16.934	12997008
⁶⁵ Cu	9.96	12844003
	13.9	12844003
⁸⁹ Y	9.954	12994003
	13.934	12994003
	16.93	12994003
⁹³ Nb	9.941	12995003
	13.915	12995003
¹²⁰ Sn	9.907	13158002
	13.894	13158002
²⁰⁸ Pb	1.8	400750051
	5.969	13531003
	6.967	13531003
	7.962	13531003
	8.958	13531003
	9.95	13531003
²⁰⁹ Bi	4.5	10855002
	6	13680002
	9	13680002

Comments

No discrepancies were identified between the KD and KDUQ corpora for neutron analyzing powers. For related information, see the Comments for neutron differential elastic cross sections above (note that the formatting of the original KD corpus did not explicitly distinguish between inclusion of differential elastic cross section data sets and analyzing power data sets).

Neutron total cross sections

Isotope	Energy	EXFOR Acc.
^{nat} Mg	5.293-297.8	13753010
	0.008-39.807	10791002
²⁷ Al	1.999-80.62	12882005
	5.293-297.772	13569008
	0.935	21926003
	0.25-19.286	22331004
^{nat} Si	0.187-47.68	10377005
	1.996-79.828	12882006
	5.293-297.772	13569009
^{nat} S	2.259-14.888	10047018
	0.102	11540002
	5.293-297.8	13753012
⁴⁰ Ca	0.04-6.058	10721002
	5.293-297.772	13569010
⁴⁸ Ti	0.974-4.025	10669002
	5.293-297.8	13753015
^{nat} Cr	0.185-29.306	10342004
⁵² Cr	1.0-4.15	12750002
	5.293-297.8	13753017
	0.021	20435003
^{nat} Fe	2.268	13727002
⁵⁶ Fe	0.187-48.608	10377007
	5.293-297.8	13753019
	1.0-18.026	22258002
⁵⁸ Ni	0.003-67.481	12972008
	5.293-297.8	13753021
	0.5-19.54	22314006
^{nat} Cu	1.2-4.5	12869002
	5.293-297.772	13569011
⁸⁹ Y	1.822-19.52	12853010
	5.293-297.8	13753022
⁹⁰ Zr	0.433-1.54	10338004
	0.933-5.054	10468002
	5.293-297.772	13569012
⁹³ Nb	2.349	13736003
	0.75-3.963	12797002
	0.215-1.32	12853018
^{nat} Mo	5.293-297.772	13569013
	2.253-14.713	10047047
	0.103-1.105	10277034
	0.215-1.32	12853021
	1.822-19.28	12853022
^{nat} Sn	0.985-4.195	12853023
	5.293-297.8	13753023
	0.002	10639005
	0.215-1.32	12853042
	5.293-297.772	13569014
^{nat} Ce	2.26-14.866	10047062
	17.5-27.8	11108033
	2.253-56.92	12891002

	0.182	20819002
	160.0-280.0	22117011
	1.013	30134005
¹⁹⁷ Au	0.048-4.389	10935003
	5.293-297.8	13753026
^{nat} Hg	2.255-14.873	10047087
	0.04	11953009
	5.293-297.8	13753027
	1.007-2.007	30134010
²⁰⁸ Pb	2.491-14.137	10047089
	0.717-1.719	12215003
	17.665	13735002
	5.293-297.772	13569018
²⁰⁹ Bi	2.376-13.977	10047093
	0.011-1.013	10449003
	1.237-3.353	108460021
	69.547	13199004
	5.293-297.772	13569019

Comments

^{nat}Mg: a data set from the keV range to 39 MeV is referenced in the KD corpus as being from Lawson et al. (ORNL Report 6420), but these data were not available on EXFOR. For the KDUQ corpus, we used what are ostensibly the same data as published by Weigmann et al. (PRC 14 (1976) 1328).

²⁷Al: The data for this nucleus we sourced from EXFOR, where they are assigned slightly different references compared to the references given in the KD corpus (the scattering energy ranges listed in EXFOR match those as shown in the KD corpus).

^{nat}Si: The data set from Larson et al. (ORNL Report No. ORNL-TM-5618 (1976)) only has data up to 0.7 MeV, making it of limited value to OMP analysis that assumes a degree of smoothness in the imaginary strength not fulfilled in the resolved-resonance region. As such, we omitted this from the KDUQ corpus. The remaining TCS data we sourced from EXFOR, where they are listed under slightly different references compared to the KD corpus.

^{nat}S: The KD corpus refers to Finlay et al. (PRC 47 (1993) 237) for this data set, but this reference has no data for natural S. We assume that the KD corpus meant to reference Abfalterer et al. (PRC 63 (2001) 44608), the other large-scale experimental campaign of neutron total cross sections conducted at LANSCE. In the KDUQ corpus, we use the data from that source.

⁴⁰Ca: the data set from Johnson et al. (Bull. Am. Phys. Soc. 23 (1978) 636) is on EXFOR listed under Int. Conf. on Nucl. Phys., Munich 1 525

(1973); we used the latter reference and data set for the KDUQ corpus.

^{52}Cr : the data set from Perey (Priv. Comm. (1973)) listed as being for ^{52}Cr in the KD corpus is actually for $^{\text{nat}}\text{Cr}$; we assigned it to $^{\text{nat}}\text{Cr}$ for the KDUQ corpus.

^{56}Fe : the data set from U.S. D.O.E. Nuclear Data Committee Reports, No.33, p.142 (1984) listed as being for ^{56}Fe is actually for $^{\text{nat}}\text{Fe}$; we assigned it to $^{\text{nat}}\text{Fe}$ for the KDUQ corpus.

^{58}Ni : EXFOR doesn't appear to contain the data set with maximum scattering energy of 67.5 MeV from Perey et al. (ORNL Report No. ORNL-TM-10841 (1988)); for the KDUQ corpus, we used the data set from Perey et al. (PRC 47 (1993) 1143).

$^{\text{nat}}\text{Cu}$: the data set from Pandey et al. (PRC 15 (1977) 600) only has data up to 1.12 MeV for ^{63}Cu and ^{65}Cu , making it of limited value to OMP analysis that assumes some degree of smoothness in the imaginary strength not fulfilled in the resonance region. As such, we omitted these from the KDUQ corpus.

$^{\text{nat}}\text{Ce}$: We sourced these data from EXFOR, where they are assigned slightly different references compared to the KD corpus.

Proton differential elastic cross sections

			35.2	O0083016
			39.8	O0456003
			49.35	O0788006
			65.0	E1201002
			156.0	O0049006
	⁵⁸ Ni		10.7	O0446002
			11.7	-
			14.4	O0446002
			15.4	O0446002
			16.0	C0893012
			17.0	O0262007
			18.6	D0286017
			21.3	O0434008
			22.2	C1019017
			30.3	O0142007
			35.2	O1198021
			39.6	O04360031
			40.0	O02080071
			61.4	O0211013
			65.0	O0032008S
			100.4	O0300003
			160.0	O0302005
			178.0	D0189002
			192.0	E1704002
	⁶⁰ Ni		14.4	O0446003
			15.4	O0446003
			16.0	C0893014
			18.6	D0286018
			30.3	O0142008
			30.8	O0157002
			39.6	O04360021
			65.0	O0032063S
	⁹⁰ Zr		9.7	O03930071
			12.7	O03890021
			14.71	O0372013
			16.0	C0893028
			18.8	O0370002
			22.5	C0085002
			30.0	D0295002
			40.0	O02080081
			49.35	O0788017
			61.4	O0211015
			65.0	O0032021S
			80.0	T0101003
			100.4	O0300004
			135.0	T0101003
			156.0	O0049009
			160.0	T0101003
	¹²⁰ Sn		9.7	O03930081
			16.0	C0893032
			20.4	O0169020
			24.6	O0169026
			30.3	O0142010
			40.0	O0328007
Isotope	Energy	EXFOR	Acc.	
²⁷ Al	17.0	O0262004		
	28.0	C1208002		
	35.2	O0083019		
	61.4	O0211003		
	142.0	O0247006		
	156.0	O0049003		
	160.0	D0283004		
	183.0	O0365008		
²⁸ Si	17.8	O0254009		
	28.0	C1208003		
	30.4	O0124002		
	40.0	O0162021		
	65.0	E0166008		
	80	C0084002		
	100	C0084002		
	135	C0084002		
	179	C0084002		
	180.0	C0152002		
⁴⁰ Ca	16.0	C0893002		
	40.0	O02080061		
	45.5	C0076002		
	61.4	O0211012		
	65.0	O0032002S		
	75.0	O0553010		
	80.0	T0101002		
	135.0	T0101002		
	152.0	O0553004		
	156.0	O0049004		
	160.0	T0101002		
	181.5	T0108005		
	201.4	C0148006		
⁵⁴ Fe	9.69	O03930021		
	12.0	C1024002		
	16.0	C0893008		
	17.2	O0091002		
	18.6	D0286015		
	19.6	O0079006		
	20.4	O0091004		
	24.6	O0091006		
	30.4	O12430031		
	35.2	O1198020		
	39.8	O0456002		
	49.35	O0788007		
	65.0	O0032061S		
^{nat} Fe	182.4	O0365012		
⁵⁶ Fe	10.93	-		
	11.7	-		
	16.0	C0893010		
	18.6	D0286016		
	19.1	O0167006		
	30.3	O0142005		

	100.4	O0300005
	156.0	O0049010
	160.0	O0479004
²⁰⁸ Pb	16.0	C0893042
	21.0	O0287011
	24.1	O0287012
	26.3	O0287013
	30.3	O0142011
	35.0	O0287015
	40.0	O02080091
	45.0	O0287016
	47.3	O0287017
	49.35	O0788009
	61.4	O0211017
	65.0	O0032025S
	80.0	T0101004
	121.0	T0101004
	156.0	O0049012
	160.0	O0302002
	182.0	T0101004
	185.0	O0287002
	200.0	C0081005
²⁰⁹ Bi	16.0	C0893044
	65.0	E0773008
	78.0	O0553014
	153.0	O0553006
	156.0	O0049013

Comments

²⁸Si: The data set listed in the KD corpus as having a scattering energy of 198.1 MeV and being from Olmer et al. (PRC 29 (1984) 361) does not appear in the referenced publication; instead, there are data sets at 179 MeV for elastic scattering and 175 MeV for analyzing powers that are listed in EXFOR for this reference. We used these latter data sets in the KDUQ corpus. The data set at 134.2 MeV and listed in the KD corpus as being from Schwandt et al. (PRC 26 (1982) 55) is instead listed in EXFOR as being from Olmer et al. (PRC 29 (1984) 361). We used this latter reference for the KDUQ corpus.

⁴⁰Ca: The data sets at 14.5, 18.6 and 21 MeV from Boschitz, Bercaw, and Vincent (Phys. Lett. 13 (1964) 322) were not available in EXFOR and apparently have not been digitized. We omitted these data sets from the KDUQ corpus.

⁵⁴Fe: The data set at 11 MeV from Benevise, Mitchel, and Fulmer (Phys. Rev. 133 (1964) B317) listed in the KD corpus as being from ⁵⁴Fe is actually from ⁵⁶Fe; it already is referenced under ⁵⁶Fe in the KD corpus. We included it as being from ⁵⁶Fe in the KDUQ corpus. The data set at 35.0

MeV from Colombo et al. (J. Phys. Soc. Jpn. 44 (1978) 543) was not available in EXFOR; for the KDUQ corpus, we used the EXFOR data set at 35.2 MeV associated with most of the same authors (PRC 21 844 (1980)).

⁵⁶Fe: The data sets at 17.2, 20.4, and 24.6 listed in the KD corpus as being from J.P.M.G. Melssen's PhD thesis (Technische Hogeschool Eindhoven, 1978) are instead listed in EXFOR as being from Van Hall et al. (NP/A 291 63 (1977)); we used this latter reference and its data sets for the KDUQ corpus. The data set listed in the KD corpus as having a scattering energy of 15.3 MeV was not available in the listed Van Hall et al. reference; as such, we omitted it from the KDUQ corpus. The data set at 14.5 MeV from Rosen et al. (Ann. Phys. (N.Y.) 34 (1965) 96) was not available in EXFOR; for the KDUQ corpus, we used the data set at 14.0 MeV from Rosen et al. (PRL 10 246 (1963)) from EXFOR. We were unable to locate the data sets at 10.93 and 11.7 MeV in EXFOR. For these data we used the same reference as listed in the KD corpus: Benveniste, Mitchell, and Fulmer (Phys. Rev. 133 B317).

⁵⁸Ni: The data set listed in the KD corpus as having a scattering energy of 17.8 MeV from Payton and Schrank (PR 101 1358 (1956)) was apparently collected at 17.0 MeV, per the original publication and EXFOR. We used the latter energy in the KDUQ corpus. The data set at 35.2 MeV listed in the KD corpus as being from Eliyakut-Roshko et al. (PRC 51 1295 (1995)) is apparently from Fabrici et al. (PRC 21 844 (1980)) and is available in EXFOR; we used the latter reference and EXFOR data sets for the KDUQ corpus. The data set at 200 MeV from Sakaguchi et al. (RCNP Annual Report, 1993, p.4) was not available in EXFOR; for the KDUQ corpus, we used EXFOR data at 192 MeV from Sakaguchi et al. (PRC 57 1749 (1998)). We were unable to locate the data set 11.7 MeV in EXFOR. For these data we used the same reference as listed in the KD corpus: Benveniste, Mitchell, and Fulmer (Phys. Rev. 133 B317).

^{nat}Ni: The data set listed in the KD corpus as having scattering energy of 17 MeV and being from Devins et al. (Nucl. Phys. 35 (1962) 617) was apparently actually collected at a scattering energy of 30.8 MeV, and using a ⁶⁰Ni target. For the KDUQ corpus, we assigned this latter energy and target to this data set.

⁶⁰Ni: The data set at 40 MeV from Blumberg et al. (PR 147 812 (1966)) listed in the KD corpus as being from ⁶⁰Ni are actually from ⁵⁸Ni; we assigned it to ⁵⁸Ni for the KDUQ corpus.

^{90}Zr : The data sets listed in the KD corpus as having scattering energies of 12.7 and 14.7 MeV do not appear in Greenlees et al. (PRC 3 1231 (1971)); for the KDUQ corpus, we used the data set at 12.7 MeV from Dickens et al. (PR 168 1355 (1968)) and at 14.7 MeV from Matsuda et al. (JPJ 22 1311 (1967)), as listed in EXFOR. The data set at 180 MeV from Nadasen et al. (PRC 23 1023 (1981)) is not available on EXFOR, only in a difficult-to-parse figure from the original paper. As such, we omitted this dataset from the KDUQ corpus.

^{208}Pb : The data sets at 11, 12, and 13 MeV from Kretschmer et al. (PLB 87 343 (1979)) are not available in EXFOR, only in a difficult-to-parse figure in the original paper. As such, we omitted these data sets from the KDUQ corpus. The data set at 100 MeV doesn't appear to originate from Nadasen et al. (PRC 23 (1981) 1023); there is a data set at 98 MeV from a related publication Schwandt et al. (PRC 26 55 (1982)) for analyzing powers. We included the latter data set in the KDUQ corpus.

^{209}Bi : The data set at 57 MeV from Yamabe et al. (JPJ 17 729 (1962)) does not include errors; we omitted it from the KDUQ corpus.

Proton analyzing powers

Isotope	Energy	EXFOR Acc.
²⁸ Si	17.8	O0254003
	65	E0166009
	80	C0084003
	100	C0084003
	135	C0084003
	175	C0084003
	180	C0152004
⁴⁰ Ca	16	C0893003
	26.3	O0490006
	30.05	O0490003
	40	O02080062
	45.5	C0076003
	65	E0166013
	75	O0553011
	80.2	T0108004
	152	O0553005
	181.5	T0108004
	201.4	C0148008
⁵⁴ Fe	9.69	O03930022
	16	C0893009
	17.2	O0091003
	18.6	D0286005
	20.4	O0091005
	24.6	O0091007
^{nat} Fe	155	O0218002
	179	D0285008
⁵⁶ Fe	14	C1100005
	17.2	O0091009
	18.6	D0286006
	20.4	O0091011
	24.6	O0091013
	65	E1201003
⁵⁸ Ni	18.6	D0286007
	20.4	O0091015
	20.9	O0434009
	24.6	O0091017
	30.04	O0490004
	40	O02080072
	65	E1201007
	178	D0189003
	192	E1704005
^{nat} Ni	155	O0218003
⁶⁰ Ni	20.4	O0091019
	24.6	O0091021
	65	E1201009
⁹⁰ Zr	9.7	O03930072
	20.25	O0392007
	30	D0295003
	40	O02080082
¹²⁰ Sn	9.7	O03930082

20.4 O0169021

24.6 O0169027

40 O0328011

²⁰⁸ Pb	26.3	O0490008
	40	O02080092
	49.35	O0788025
	65	E1201015
	79.8	T0108010
	98	T0108010
	182	T0108010
	200	C0081006

Comments

No discrepancies were identified between the KD and KDUQ corpora for proton analyzing powers. For related information, see the Comments for proton differential elastic cross sections above (note that the formatting of the original KD corpus did not explicitly distinguish between inclusion of differential elastic cross section data sets and analyzing power data sets).

Proton reaction cross sections

Isotope	Energy	EXFOR Acc.
²⁷ Al	8.87-10.39	D0314002
	8.8	D0535002
	9.89	C1042003
	9.9-10.12	O0741004
	16.29	O0368004
	24.8-46.3	O0325003
	29	O0369003
	34	O0732003
	40-60.8	O0081008
	61	O0475003
	77-133	C1211003
	99.7	O0340006
	134	O1947003
	179.6	D0533005
	185	-
	234	O0213004
^{nat} Si	20.7-38.1	C1862005
	24.7-47.8	O0325006
	65.5	O0579005
⁴⁰ Ca	10.34-21.59	O0341003
	24.9-48	O0330003
	28.5	O0150002
	65.5	O0579006
^{nat} Ca	99.3	O0340007
	179.6	D0533006
^{nat} Fe	8.8	D0535004
	9.21-11.25	D0314003
	9.89	C1042006
	9.97-10.2	O0741013
	15.8	C1864004
	28	C1212007
	34	O0732004
	61	O0475004
	98.7	O0340011
	179.6	D0533007
	230	O0213005
⁵⁶ Fe	14.5	C1217004
	20.8-47.8	T0100004
	28.5	O0150003
	40-60.8	O0081013
⁶³ Cu	6.75	-
	8.7	D0535009
	9.1	D0534004
	9.11-11.18	D0314010
	9.85	-
	14.5	C1217010
^{nat} Cu	8.78-11.21	D0337003
	8.8	D0535008
	8.9	C0067002
	9.05	D0534006

	9.3	O1948003
	9.9-10.12	O0741006
	15.8	C1864003
	16.37	O0368006
	28	C1212016
	77-133	C1211004
	99	O0340014
	134	O1947004
	185	-
	225	O0213006
⁹⁰ Zr	14.5	C1217014
	30-60.8	O0081004
^{nat} Zr	9.2	D0534007
	10.03-10.25	O0741016
	98.8	O0340016
^{nat} Sn	9.99-10.21	O0741015
	34	O0732005
	61	O0475005
	99.1	O0340023
	221	O0213009
	14.5	C1217022
¹²⁰ Sn	22.8-47.9	C0424006
	28.5	O0150007
	30-49.5	O0081005
	65.5	O0579012
^{nat} Pb	9.92	O0741019
	16.31	O0368007
	34	O0732006
	61	O0475006
	77-133	C1211006
	99.2	O0340040
	134	O1947006
	185	-
²⁰⁸ Pb	226	O0213010
	21.1-48	O0330004
	28.5	O0150008
	30-60.8	O0081007
	65.5	O0579014

Comments

²⁷Al: we were unable to locate the datum at 185 MeV in EXFOR. For this datum we used the same reference as listed in the KD corpus: Millburn et al. (Phys. Rev. 95 1268 (1954)).

⁴⁰Ca: the data from 99.3 and 179.6 MeV listed as being for ⁴⁰Ca in the KD corpus are actually from ^{nat}Ca; we assigned them to ^{nat}Ca for the KDUQ corpus.

⁵⁶Fe: many of the data listed as being for ⁵⁶Fe in the KD Corpus are actually for ^{nat}Fe; we assigned them to ^{nat}Fe for the KDUQ corpus.

⁶³Cu: many of the data listed as being for ⁶³Cu in

the KD Corpus are actually for $^{\text{nat}}\text{Cu}$; we assigned them to $^{\text{nat}}\text{Cu}$ for the KDUQ corpus. We were unable to locate the data at 6.75 and 9.85 MeV in EXFOR. For these data we used the same references as listed in the KD corpus: Dell, Ploughe, and Hausman (Nucl. Phys. 64 (1965) p. 513), and Albert and Hansen (PRL 6 (1961) p. 13), respectively.

$^{\text{nat}}\text{Cu}$: we were unable to locate the datum at 185 MeV in EXFOR. For this datum we used the same reference as listed in the KD corpus: Millburn et al. (Phys. Rev. 95 1268 (1954)).

^{90}Zr : several of the data listed as being for ^{90}Zr in the KD Corpus are actually for $^{\text{nat}}\text{Zr}$; we assigned them to $^{\text{nat}}\text{Zr}$ for the KDUQ corpus.

^{120}Sn : several of the data listed as being for ^{120}Sn in the KD Corpus are actually for $^{\text{nat}}\text{Sn}$; we assigned them to $^{\text{nat}}\text{Sn}$ for the KDUQ corpus.

$^{\text{nat}}\text{Pb}$: we were unable to locate the datum at 185 MeV in EXFOR. For this datum we used the same reference as listed in the KD corpus: Millburn et al. (Phys. Rev. 95 1268 (1954)).

^{208}Pb : several of the data listed as being for ^{208}Pb in the KD Corpus are actually for $^{\text{nat}}\text{Pb}$; we assigned them to $^{\text{nat}}\text{Pb}$ for the KDUQ corpus.

CHUQ CORPUS

Neutron differential elastic cross sections

Isotope	Energies	EXFOR Acc. #
^{40}Ca	13.905	12996002
	16.916	12996002
$^{\text{nat}}\text{Ca}$	11.01	10633005
^{51}V	11.01	10633006
^{54}Fe	9.94	10958002
	11.0	12862002
	13.92	10958002
	20.0	12862002
	22.0	12862002
	24.0	12862002
	26.0	12862002
^{55}Mn	11.01	10633007
^{56}Fe	11.0	12862003
	20.0	12862003
	26.0	12862003
^{58}Ni	9.958	12930002
	13.941	12930002
^{59}Co	11.01	10633009
^{60}Ni	9.958	12930004
	13.941	12930004
^{65}Cu	9.94	10958008
	13.92	10958008
^{88}Sr	11.0	10729004
^{89}Y	11.0	12774002
	11.0	12790002
^{90}Zr	11.0	10729002
^{92}Mo	11.0	10729007
	11.0	10867002
	11.01	10633012
	20.0	10867002
	26.0	10867002
^{93}Nb	11.01	10633011
^{96}Mo	11.0	10867003
	11.01	10633013
	20.0	10867003
	26.0	10867003
^{98}Mo	11.0	10867004
	11.01	10633014
	20.0	10867004
	26.0	10867004
^{100}Mo	11.0	10867005
	11.01	10633015
	20.0	10867005
	26.0	10867005
^{116}Sn	9.945	13158005
	11.0	10817006
	13.925	13158005
	24.0	10817006

^{118}Sn	11.0	10817007
	24.0	10817007
^{120}Sn	9.943	13158007
	11.0	10817008
	11.01	10633017
	13.923	13158007
^{122}Sn	16.905	13158007
^{124}Sn	11.0	10817010
	24.0	10817010
^{165}Hg	11.01	10633018
^{206}Pb	11.01	10633020
^{208}Pb	11.0	10871002
	13.9	13685002
	16.9	13685002
	20.0	10871002
	26.0	10871002
^{209}Bi	11.01	10633022

Comments

$^{\text{nat}}\text{Ca}$, ^{51}V , ^{55}Mn , ^{59}Co , ^{93}Nb , ^{92}Mo , ^{96}Mo , ^{98}Mo , ^{100}Mo , ^{120}Sn , ^{165}Hg , ^{206}Pb , ^{209}Bi : as listed in EXFOR, the data sets for these nuclei from the publication of Ferrer et al. (Nucl. Phys. A 275, p. 325 (1977)) had both a DATA-ERR1 column containing a partial error in absolute units, and a DATA-ERR2 column containing a separate partial error in percent. It was unclear (to us) the relationship between these sources of error, and we were unable to determine how these data sets were treated in the canonical CH89 fit. For these data sets, we converted the percent error to absolute units and summed both sources of partial error in quadrature to yield an overall error, which was then used in the CHUQ corpus.

^{118}Sn : For the data sets at 11 and 24 MeV Rapaport et al. (Nucl. Phys. A 341 p. 56 (1980)) listed on EXFOR, a “null” value was listed in the DATA-ERR column; we elected to use the following column ERR-T, which listed a 5% relative uncertainty for all data points, for the error of these data in the CHUQ corpus.

^{54}Fe , ^{56}Fe : For the data sets at 11, 20, 22, 24, and 26 MeV by Mellema et al. (Phys. Rev. C 28 p. 2267 (1983)) listed in EXFOR, the first and second DATA-ERR columns listed errors in percent and in absolute units, respectively. The first column contained mostly “null” values. To combine the data from both columns for use in the CHUQ corpus, we converted the percent errors that did exist in the first DATA-ERR column into absolute units, then merged the two columns into one absolute DATA-ERR column.

Neutron analyzing powers

Isotope	Energies	EXFOR Acc. #
⁴⁰ Ca	10.935	12996004
	13.904	12996004
	16.923	12996004
⁵⁸ Ni	9.92	12930010
	13.91	12930010
¹¹⁶ Sn	9.907	13158002
	13.894	13158002
¹²⁰ Sn	9.906	13158003
	13.894	13158003
²⁰⁸ Pb	9.97	12844007
	13.9	12844007

Comments

⁶⁴Ni: In the data set at 65 MeV by Sakaguchi et al. (Memoirs Faculty of Sci., Kyoto Univ. Ser. Phys. 36 p.305 (1983)) as listed in EXFOR, the first datum possessed an unphysical error of 3.0034 (the analyzing power can only assume a value between -1 and 1). As this is likely a transcription or tabulation error, for the CHUQ corpus we changed this to 0.0034, which is consistent with the other listed errors from this data set.

Proton differential elastic cross sections

Isotope	Energies	EXFOR Acc. #
⁴⁰ Ca	40.0	O0328003
	65.0	O0032002S
⁴⁴ Ca	65.0	O0032056S
⁴⁶ Ti	65.0	O0032058S
⁴⁸ Ca	65.0	O0032057S
⁴⁸ Ti	16.0	C0893004
	65.0	O0032059S
⁵⁰ Ti	16.0	C0893006
	65.0	O0032060S
⁵⁴ Fe	16.0	C0893008
	17.2	O0091002
	20.4	O0091004
	24.6	O0091006
	40.0	O0162023
	65.0	O0032061S
⁵⁶ Fe	17.2	O0091008
	20.4	O0091010
	24.6	O0091012
	65.0	O0032062S
⁵⁸ Ni	16.0	C0893012
	20.4	O0091014
	24.6	O0091016
	40.0	O0162025
	65.0	O0032008S
⁵⁹ Co	40.0	O0328004
	65.0	O0032015S
⁶⁰ Ni	16.0	C0893014
	20.4	O0091018
	24.6	O0091020
	40.0	O0162027
	65.0	O0032063S
⁶² Ni	20.4	O0091022
	24.6	O0091024
	65.0	O0032064S
⁶³ Cu	16.0	C0893016
⁶⁴ Ni	20.4	O0169004
	65.0	O0032065S
⁶⁴ Zn	20.4	O1109002
⁶⁵ Cu	16.0	C0893018
⁶⁶ Zn	20.4	O1109004
⁶⁸ Zn	20.4	O1109006
	40.0	O0328005
⁷⁰ Zn	20.4	O1109008
⁷² Ge	22.3	O1103004
⁷⁴ Ge	22.3	O1103006
⁷⁶ Se	16.0	C0893020
⁷⁸ Se	16.0	C0893022
	22.3	O1103012
⁸⁰ Se	16.0	C0893024
	22.3	O1103014

⁸² Se	16.0	C0893026
⁸⁶ Sr	24.6	O0169006
⁸⁸ Sr	24.6	O0169008
⁸⁹ Y	65.0	O0032066S
⁹⁰ Zr	16.0	C0893028
	40.0	O0328006
	65.0	O0032021S
⁹⁸ Mo	65.0	O0032067S
¹⁰⁰ Mo	65.0	O0032068S
¹⁰⁶ Cd	22.3	O1104002
¹⁰⁸ Cd	22.3	O1104004
¹¹⁰ Cd	20.4	O0169010
	22.3	O1104006
¹¹² Cd	20.4	O0169012
	22.3	O1104008
¹¹⁴ Cd	20.4	O0169014
	22.3	O1104010
¹¹⁶ Cd	22.3	O1104012
¹¹⁶ Sn	16.0	C0893030
	20.4	O0169016
¹¹⁸ Sn	20.4	O0169018
¹²⁰ Sn	16.0	C0893032
	20.4	O0169020
	24.6	O0169026
	40.0	O0328007
¹²² Sn	20.4	O0169022
¹²⁴ Sn	16.0	C0893034
	20.4	O0169024
¹³⁴ Ba	16.0	C0893036
¹³⁶ Ba	16.0	C0893038
¹³⁸ Ba	16.0	C0893040
¹⁴⁴ Sm	65.0	O0032069S
²⁰⁸ Pb	40.0	O0328008
	65.0	O0032025S
²⁰⁹ Bi	65.0	O0032070S

Comments

¹³⁴Ba, ¹³⁶Ba: the data sets for these nuclei from Varner, PhD thesis (1986) are listed in the CH89 Corpus as being from ¹³²Ba and ¹³⁴Ba, but instead listed in EXFOR as being from ¹³⁴Ba and ¹³⁶Ba. As ¹³²Ba is 0.1% naturally abundant, it is highly unlikely this was actually used as a target. For the CHUQ corpus, we assigned the data sets to ¹³⁴Ba and ¹³⁶Ba, respectively.

⁵⁸Ni: we were unable to locate the data set listed in the CH89 corpus as having a scattering energy of 27.2 MeV. This scattering energy was not reported (to our knowledge) in any of the references cited in the CH89 corpus for the Eindhoven (p,p) scattering data sets.

^{98}Mo : in the data set from Sakaguchi et al. (Memoirs Faculty of Sci., Kyoto Univ. Ser. Phys. 36 p.305 (1983)), the cross section value recorded in EXFOR at an angle of 58.03 degrees is anomalously lower than the adjacent cross sections reported at 55.52 and 60.54 degrees, possibly due to a tabulation error. For the CHUQ corpus we retained the anomalous value.

^{142}Nd : we were unable to locate the data set listed in the CH89 corpus with a scattering energy of 17.2 MeV. To our knowledge, no data for proton scattering on ^{142}Nd are available in EXFOR, nor is ^{142}Nd listed as a target under any of the references cited in the CH89 corpus for the Eindhoven (p,p) scattering data sets.

^{148}Sm , ^{150}Sm : we were unable to locate the data sets listed in the CH89 corpus with a scattering energy of 20.4 MeV. To our knowledge, no data for proton scattering on Sm isotopes at or around 20 MeV are available in EXFOR, nor are any Sm isotopes listed as a target under any of the references cited in the CH89 corpus for the Eindhoven (p,p) scattering data sets.

^{40}Ca , ^{54}Fe , ^{58}Ni , ^{60}Ni , ^{59}Co , ^{68}Zn , ^{90}Zr , ^{120}Sn , ^{208}Pb : as listed in EXFOR, the data sets for these nuclei, associated with two publications by Fricke et al. (Phys. Rev. 163 p.1153 (1967); Phys. Rev. 156 p.1207 (1967)) have mostly null values in the ERR-T column. It was unclear to us how to combine or assess the few given ERR-T data with the error data in the other provided error columns. As such, we ignored the ERR-T column in these cases and took the DATA-ERR column for the total error.

Proton analyzing powers

Isotope	Energies	EXFOR Acc. #
⁴⁰ Ca	65.0	O0032002A
⁴⁴ Ca	65.0	O0032056A
⁴⁶ Ti	65.0	O0032058A
⁴⁸ Ca	65.0	O0032057A
⁴⁸ Ti	16.0	C0893005
	65.0	O0032059A
⁵⁰ Ti	16.0	C0893007
	65.0	O0032060A
⁵⁴ Fe	16.0	C0893009
	17.2	O0091003
	20.4	O0091005
	24.6	O0091007
	40.0	O0162024
	65.0	O0032061A
⁵⁶ Fe	17.2	O0091009
	20.4	O0091011
	24.6	O0091013
	65.0	O0032062A
⁵⁸ Ni	16.0	C0893013
	20.4	O0091015
	24.6	O0091017
	40.0	O0162026
	65.0	O0032008A
⁵⁹ Co	40.0	O0328009
	65.0	O0032015A
⁶⁰ Ni	16.0	C0893015
	20.4	O0091019
	24.6	O0091021
	40.0	O0162028
	65.0	O0032063A
⁶² Ni	20.4	O0091023
	24.6	O0091025
	65.0	O0032064A
⁶³ Cu	16.0	C0893017
⁶⁴ Ni	20.4	O0169005
	65.0	O0032065A
⁶⁴ Zn	20.4	O1109003
⁶⁵ Cu	16.0	C0893019
⁶⁶ Zn	20.4	O1109005
⁶⁸ Zn	20.4	O1109007
	40.0	O0328010
⁷⁰ Zn	20.4	O1109009
⁷² Ge	22.3	O1103005
⁷⁴ Ge	22.3	O1103007
⁷⁶ Se	16.0	C0893021
⁷⁸ Se	16.0	C0893023
	22.3	O1103013
⁸⁰ Se	16.0	C0893025
	22.3	O1103015
⁸² Se	16.0	C0893027

⁸⁶ Sr	24.6	O0169007
⁸⁸ Sr	24.6	O0169009
⁸⁹ Y	65.0	O0032066A
⁹⁰ Zr	16.0	C0893029
	65.0	O0032021A
⁹⁸ Mo	65.0	O0032067A
¹⁰⁰ Mo	65.0	O0032068A
¹⁰⁶ Cd	22.3	O1104003
¹⁰⁸ Cd	22.3	O1104005
¹¹⁰ Cd	20.4	O0169011
	22.3	O1104007
¹¹² Cd	20.4	O0169013
	22.3	O1104009
¹¹⁴ Cd	20.4	O0169015
	22.3	O1104011
¹¹⁶ Cd	22.3	O1104013
¹¹⁶ Sn	16.0	C0893031
	20.4	O0169017
¹¹⁸ Sn	20.4	O0169019
¹²⁰ Sn	16.0	C0893033
	20.4	O0169021
	24.6	O0169027
	40.0	O0328011
¹²² Sn	20.4	O0169023
¹²⁴ Sn	16.0	C0893035
	20.4	O0169025
¹³⁴ Ba	16.0	C0893037
¹³⁶ Ba	16.0	C0893039
¹³⁸ Ba	16.0	C0893041
¹⁴⁴ Sm	65.0	O0032069A
²⁰⁸ Pb	65.0	O0032025A
²⁰⁹ Bi	65.0	O0032070A

Comments

¹³⁴Ba, ¹³⁶Ba: the data sets for these nuclei from Varner, PhD thesis (1986) are listed in the CH89 Corpus as being from ¹³²Ba and ¹³⁴Ba, but instead listed in EXFOR as being from ¹³⁴Ba and ¹³⁶Ba. As ¹³²Ba is 0.1% naturally abundant, it is highly unlikely this was actually used as a target. For the CHUQ corpus, we assigned the data sets to ¹³⁴Ba and ¹³⁶Ba, respectively.

⁵⁸Ni: we were unable to locate the data set listed in the CH89 corpus with a scattering energy of 27.2 MeV. This scattering energy was not reported (to our knowledge) in any of the references cited in the CH89 corpus for the Eindhoven (p,p) scattering data sets.

¹⁴²Nd: we were unable to locate the data set listed in the CH89 corpus with a scattering energy of 17.2 MeV. To our knowledge, no data for proton scattering on ¹⁴²Nd are available in EXFOR, nor is

^{142}Nd listed as a target under any of the references cited in the CH89 corpus for the Eindhoven (p,p) scattering data sets.

^{148}Sm , ^{150}Sm : we were unable to locate the data sets listed in the CH89 corpus with a scattering energy of 20.4 MeV. To our knowledge, no data for proton scattering on Sm isotopes at or around 20 MeV are available in EXFOR, nor are any Sm isotopes listed as a target under any of the references cited in the CH89 corpus for the Eindhoven (p,p) scattering data sets.

^{54}Fe , ^{58}Ni , ^{60}Ni , ^{64}Ni , ^{59}Co , ^{64}Zn , ^{66}Zn , ^{68}Zn , ^{70}Zn , ^{72}Ge , ^{74}Ge , ^{78}Se , ^{80}Se , ^{86}Sr , ^{88}Sr , ^{110}Cd , ^{112}Cd , ^{114}Cd , ^{116}Sn , ^{118}Sn , ^{120}Sn , ^{122}Sn , ^{124}Sn : as listed in EXFOR, the data sets for these nuclei, associated with two publications by Fricke et al. (Phys. Rev. 163 p.1153 (1967); Phys. Rev. 156 p.1207 (1967)), one publication by Moonen et al. (J. Phys. G 19, p.635 (1993)), and one publication by Wassenaar et al. (J. Phys. G 15, p.181 (1989)) have mostly (or many) null values in the ERR-T column. It was unclear to us how to combine or assess the given ERR-T data with the error data in the other provided error columns. As such, we ignored the ERR-T column in these cases and took the DATA-ERR column for the total error.

TEST CORPUS

Neutron differential elastic cross sections

Isotope	Energies	EXFOR Acc.
²⁷ Al	15.431	13903002
²⁸ Si	15.43	14345002
	18.9	14345002
³² S	7.96	14345004
	9.95	14345004
	11.93	14345004
	13.92	14345004
	15.44	14345004
	16.92	14345004
	18.9	14345004
^{nat} Ar	6.0	14371004
⁴⁰ Ca	11.9	14303002
	16.9	14303003
	65.0	13946003
	75.0	13946003
	85.0	13946003
	95.0	13946003
	107.5	13946003
	127.5	13946003
	155.0	13946003
	185.0	13946003
	225.0	13946003
⁴⁸ Ca	11.9	14303004
	16.8	14303005
⁵⁴ Fe	2.0	14451008
	2.25	14451008
	2.5	14451008
	2.75	14451008
	3.0	14451008
	3.5	14451008
	4.0	14451008
	5.0	14451008
	6.0	14451008
^{nat} Fe	1.75	14451003
	8.17	32673002
⁵⁶ Fe	1.3	14462002
	1.5	14462002
	1.8	14462002
	2.0	14462002
	2.25	14462002
	2.5	14462002
	2.75	14462002
	3.0	14462002
	3.5	14462002
	4.0	14462002
	4.5	14462002
	4.9	14462002
	5.94	14462002
	6.96	14462002

	7.96	14462002
	96.0	22987002
	96.0	22987005
	96.0	23059003
⁵⁹ Co	9.953	13903004
	11.944	13903004
	15.425	13903004
	16.879	13903004
	18.862	13903004
^{nat} Cu	6.95	22974002
	8.07	22974002
	8.96	22974002
	9.9	22974002
	10.86	22974002
	11.9	22974002
	12.93	22974002
	14.18	22974002
⁸⁹ Y	96.0	22987003
¹¹² Sn	11.0	14662002
	17.0	14662003
¹²⁴ Sn	11.0	14662004
	17.0	14662005
^{nat} Gd	0.334	13894002
	0.456	13894002
	0.55	13894002
	0.649	13894002
	0.785	13894002
	0.919	13894002
	1.08	13894002
	1.264	13894002
	1.432	13894002
	4.51	13894004
	5.01	13894004
	5.51	13894004
	5.91	13894004
	6.51	13894004
	7.14	13894004
	7.51	13894004
	8.03	13894004
	8.41	13894004
	9.06	13894004
	9.51	13894004
	9.99	13894004
¹⁸¹ Ta	0.323	13965002
	0.37	13965002
	0.414	13965002
	0.469	13965002
	0.522	13965002
	0.568	13965002
	0.615	13965002
	0.666	13965002
	0.719	13965002
	0.767	13965002
	0.815	13965002

0.867	13965002		9.06	13878004
0.911	13965002		9.51	13878004
0.967	13965002		9.99	13878004
1.015	13965002	¹⁹⁷ Au	4.51	14033002
1.068	13965002		5.01	14033002
1.11	13965002		5.51	14033002
1.163	13965002		5.91	14033002
1.213	13965002		6.51	14033002
1.261	13965002		7.14	14033002
1.309	13965002		7.51	14033002
1.359	13965002		8.03	14033002
1.415	13965002		8.41	14033002
1.465	13965002		9.06	14033002
4.51	13965002		9.51	14033002
5.01	13965002		9.99	14033002
5.51	13965002	^{nat} Pb	2.24	23156002
5.91	13965002		2.71	23156003
6.51	13965002		2.9	31687002
7.14	13965002		2.94	23156004
7.51	13965002		3.0	31687002
8.03	13965002		3.1	31687002
8.41	13965002		3.2	31687002
9.06	13965002		3.4	31687002
9.51	13965002		4.02	23156005
9.99	13965002	²⁰⁸ Pb	30.4	14317002
^{nat} W	7.19	22962002	40.0	14317003
	8.08	22962002	65.0	13946004
	9.08	22962002	75.0	13946004
	10.11	22962002	85.0	13946004
	11.04	22962002	95.0	13946004
	12.12	22962002	96.0	22987004
	13.1	22962002	96.0	22987007
	14.1	22962002	96.0	23059002
^{nat} Re	0.352	13878002	107.5	13946004
	0.446	13878002	127.5	13946004
	0.544	13878002	155.0	13946004
	0.643	13878002	185.0	13946004
	0.748	13878002	225.0	13946004
	0.846	13878002	²⁰⁹ Bi	3.99
	0.944	13878002		23156006
	1.034	13878002		
	1.144	13878002		
	1.241	13878002		
	1.345	13878002		
	1.44	13878002		
	4.51	13878004		
	5.01	13878004		
	5.51	13878004		
	5.91	13878004		
	6.51	13878004		
	7.14	13878004		
	7.51	13878004		
	8.03	13878004		
	8.41	13878004		

Comments

^{nat}Fe: We did not include the data set of E. Pirovano et al. (Phys. Rev. C 99 (2019) 024601), as it included over 100 scattering energies but only a few angles for each energy.

⁵⁴Fe: We reduced the cross section values and reported errors of J. R. Vanhoy et al. (Nucl. Phys. A 972 (2018) 107) by a factor of 1000; the data as listed in EXFOR appear to be 1000 too small, possible due to a units mismatch.

^{nat}W: We combined the data set of 0-degree elastic scattering of Schmidt et al. (EXFOR Acc. No.

22962024) with the non-0-degree elastic scattering data at the same energies by the same authors (EXFOR Acc. No. 22962002), to avoid duplicate calculations/figures of these data while using the Test corpus.

⁸⁹Y: We combined the data set of 0-degree elastic scattering of Oehr et al. (EXFOR Acc. No. 22987006) with the non-0-degree elastic scattering data at the same energies by the same authors (EXFOR Acc. No. 22987003), to avoid duplicate calculations/figures of these data while using the Test corpus.

Neutron analyzing powers

Isotope	Energies	EXFOR Acc.
²⁷ Al	15.425	13903003
²⁸ Si	15.4	14345003
	18.6	14345003
³² S	9.9	14345005
	13.9	14345005
	15.4	14345005
	16.9	14345005
⁵⁹ Co	15.273	13903005

Comments

No special cases were encountered in assembling neutron analyzing powers for the Test corpus.

Neutron total cross sections

Isotope	Energies	EXFOR Acc.
^{nat} S	14.1	31807004
⁴⁰ Ca	12.04-276.13	14269004
^{nat} Ti	0.4-24.75	14576004
	0.401-24.69	14576005
⁴⁸ Ca	12.04-276.13	14269003
⁵⁸ Ni	2.505-290.209	14661007
^{nat} Ni	2.505-290.209	14661008
⁶⁴ Ni	2.505-290.209	14661006
^{nat} Zr	0.4-24.74	14576008
	0.4-24.74	14576009
¹⁰³ Rh	2.505-290.209	14661005
¹¹² Sn	3.006-299.233	14661004
^{nat} In	14.1	31807005
^{nat} Sn	3.006-299.233	14661002
¹²⁴ Sn	3.006-299.233	14661003
^{nat} Te	14.1	31807006
^{nat} Ta	0.2-9.121	23199004
	0.732-1.853	30831002
¹⁸¹ Ta	0.4-24.72	14576002
	0.4-24.72	14576003
¹⁸² W	5.48-299.34	13887002
¹⁸⁴ W	5.48-299.34	13887003
¹⁸⁶ W	5.48-299.34	13887004
¹⁹⁷ Au	0.2-9.121	23199002
²⁰⁴ Pb	26.993-26.993	13907002
^{nat} Pb	2.505-290.209	14661015
	14.1	31807007
²⁰⁹ Bi	0.682	30831003

Comments

⁴⁰Ar: We did not include the data set of B. Bhandari et al. (Phys. Rev. Lett. 123 (2019) 042502) because the very large reported energy uncertainties made the data unsuitable for comparison against OMP predictions. For example, the reported scattering energy uncertainty for one cross section datum was approximately ± 50 MeV.

¹⁹⁷Au, ^{nat}W, ^{nat}Fe: We did not include the data sets of R. Beyer et al. (Eur. Phys. J. A 54 (2018) 81), as the scattering energies used were only up to 8 MeV, making the data only marginally useful for comparison against an optical potential.

^{nat}Fe: We did not include the data set of G. D. Kim et al. (J. Rad. and Nucl. Chem. 271 (2007) 541), as the scattering energies used were from 1.1 to 2 MeV, making the data only marginally useful for comparison against an optical potential.

¹²⁷I: We did not include the data set of G. Noguere et al. (Priv. Comm: Noguere (2005)), as the scattering energies used were from a few keV to less than 2 MeV, making the data only marginally useful for comparison against an optical potential.

^{nat}Pb: We did not include the data set of A. B. Laptev (Thesis: Laptev (2004)); the data appear not to have been corrected for dead time and are inconsistent with many other measurements of the same quantity.

Proton differential elastic cross sections

Isotope	Energies	EXFOR Acc.
²⁴ Mg	7.4	F1277002
⁵⁸ Ni	172.0	D0350003
	250.0	E2042003
⁶⁴ Zn	24.0	O2165003
¹¹⁶ Sn	295.0	E2098002
¹¹⁸ Sn	295.0	E2098003
¹²⁰ Sn	200.0	E2042007
	250.0	E2042010
	295.0	E2098004
	300.0	E2042012
¹²² Sn	295.0	E2098005
¹²⁴ Sn	295.0	E2098006

Comments

²⁷Al: we did not include the data sets of Zdravko Siketic et al. (Nucl. Instrum. Meth. 261 (2007) 414) and K. Shahzad et al. (Nucl. Sci. Techn. (Shanghai) 27 (2016) 33), as they included over 100 scattering energies but only a few angles for each energy. We also did not include the data set of S. T. Pittman et al. (Phys. Rev. C 85 (2012) 065804), as it had arbitrary (non-normalized) units.

⁴⁴Ca: we did not include the data set of S. J. Lokitz et al. (Phys. Lett. B, 599, (2004) 223), as it included over 100 scattering energies but only a few angles for each energy.

^{nat}K: we did not include the data set of M. Kokkoris et al. (Nucl. Instrum. Meth. 268 (2010) 1797), as it included over 100 scattering energies but only a few angles for each energy.

^{nat}Mg: we did not include the data set of Xiaodong Zhang et al. (Nucl. Instrum. Meth. 201 (2003) 551), as it included over 100 scattering energies but only a few angles for each energy.

⁵⁸Ni, ²⁰⁴Pb, ²⁰⁶Pb, ²⁰⁸Pb: we did not include the data sets of J. Zenihiro et al. (Phys. Rev. C 82 (2010) 044611), as they had no associated cross section errors.

⁴⁵Sc: we did not include the data set of G. Provatas et al. (Nucl. Instrum. Meth. 269 (2011) 2994), as it included over 100 scattering energies but only a few angles for each energy.

^{nat}Si: we did not include the data set of L. Csedreki et al. (Nucl. Instrum. Meth. 443 (2019) 48), as it included over 100 scattering energies but only a few angles for each energy. We did not include the data sets of Becker (Private communication (2008)), as

these elastic scattering data are dominated by resonances, making them categorically distinct from the other cross section and analyzing power data considered in this work.

¹¹⁶Sn: There were several null values in the data set of S. Terashima et al. (Phys. Rev. C 77 (2008) 024317) for the “flag” column in the EXFOR data file; we mapped these to zero, to silence the parsing error into JSON, and included the full data set for analysis.

^{nat}Ti: There were several null values in the data set of P. Hu et al. (Nucl. Instrum. Meth. 217 (2004) 551) for the “flag” column in the EXFOR data file; we mapped these to zero, to silence the parsing error into JSON, and included the full data set for analysis.

^{nat}V: We did not include the data set of X. Zhang et al. (J. Rad. Nucl. Chem. 266, (2005) 149), as it included over 100 scattering energies but only a few angles for each energy.

⁵²Cr: We did not include the data set of K. G. Leach et al. (Phys. Rev. C 100 (2019) 014320); the data as listed in EXFOR are likely subject to a tabulation error and do not correspond with the values in the referenced publication.

⁶⁴Zn: We reduced the cross section values and reported errors of K. G. Leach et al. (Phys. Rev. C 100 (2019) 014320) by a factor of 1000; the data as listed in EXFOR appear to be a factor of 1000 too large, possibly due to a units mismatch.

Proton analyzing powers

Isotope	Energies	EXFOR Acc.
^{58}Ni	172.0	D0350004
	250.0	E2042004
	295.0	E2291009
^{116}Sn	295.0	E2098007
^{118}Sn	295.0	E2098008
^{120}Sn	200.0	E2042008
	250.0	E2042011
	295.0	E2098009
	300.0	E2042013
^{122}Sn	295.0	E2098010
^{124}Sn	295.0	E2098011
^{204}Pb	295.0	E2291006
^{206}Pb	295.0	E2291007
^{208}Pb	295.0	E2291008

Comments

No special cases were encountered in assembling proton analyzing powers for the Test corpus.

Proton reaction cross sections

Isotope	Energies	EXFOR Acc.
⁴⁰ Ca	81.0-180.0	D0356003
⁵⁸ Ni	81.0	D0356004
⁹⁰ Zr	81.0-180.0	D0356005
²⁰⁸ Pb	81.0-180.0	D0356006

Comments

No special cases were encountered in assembling proton reaction cross sections for the Test corpus.